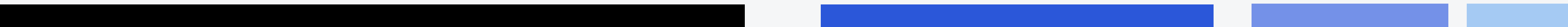


Practice Session



Question-1

The **Lahore Airport Authority** wants every Aircraft object created in the system to share a single counter that tracks the total number of aircraft registered across the whole fleet. **Suggest a solution.**

```
class Aircraft {  
};
```

Question-2

Flight class has a static data member static int activeFlights;. From main(), without creating any Flight object, you want to print the current number of active flights. **How will you do that?**

```
class Flight {  
    static int activeFlights;  
public:  
    // some function here  
};  
  
int Flight::activeFlights = 0;
```

Question-3

What is missing here?

```
class Flight {  
    int number;  
    float cost;  
};  
  
int main() {  
    int num = 1;  
    Flight flight(num);  
}
```

Question-4

What is missing here?

```
class Flight {  
    int number;  
    float cost;  
};  
  
int main() {  
    int num = 1;  
    float cost = 100f;  
    Flight flight(num, cost);}
```


Question-5

What is missing here?

```
class Flight {  
    int number;  
    float cost;  
};  
  
int main() {  
    int num = 1;  
    Flight flight = num;  
}
```

Question-6

The Flight class stores string origin; and string destination; Write a member function **Bool isSameRoute(Flight other);** that returns true only if this and the passed-in Flight object share both the same origin and the same destination.

```
class Flight {  
    string origin;  
    string destination;  
};
```

Question-7

Output?

```
class Runway {
    int length;
    public:
    Runway(int l) : length(l) {}
    Runway extend(int extra) {
        return Runway(length + extra);}
    int getLength() { return length; }
};

int main() {
    Runway r1(3000);
    _____= r1.extend(500);
    cout << r2.getLength();}
```


Question-8

You have two classes: Passenger and BoardingPass.

Write a member function inside Passenger:

BoardingPass issueBoardingPass(int seatNum, string flightCode);
that constructs and returns a BoardingPass object for this passenger.

```
class BoardingPass {
public:
    string passengerName;
    int seatNumber;
    string flightCode;
    BoardingPass(string n, int s, string f)
        : passengerName(n), seatNumber(s){};
};

class Passenger {
    string name;
public:
    Passenger(string n) : name(n) {}
    // Write the function below};
```

Question-9

Which of the following correctly accesses the getModel() member of plane?

- A. plane.getModel();
- B. plane->getModel();
- C. (*plane).getModel();
- D. Both B and C

```
Aircraft* plane = new Aircraft("A380");
```